



Embedded Automation for Smart Modular Waste Conversion



A PROJECT REPORT

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ABSTRACT

The improper segregation of waste into degradable and non-degradable categories remains a significant challenge in modern society, leading to environmental pollution and inefficient waste management. This project aims to promote effective waste segregation at the household level through the use of smart bins integrated with advanced technologies. With the rapid development of the Internet of Things (IoT) and Artificial Intelligence (AI), traditional waste management systems can be transformed into intelligent and automated solutions. The proposed system incorporates smart bins equipped with sensors capable of identifying and classifying waste materials. These bins assist users in disposing of waste correctly by automatically segregating biodegradable and non-biodegradable materials. Additionally, a 24/7 monitoring system is implemented to track the status of dumpsters in real time, ensuring timely waste collection and preventing overflow.

The project involves the design and development of a hardware prototype integrated with sensors and microcontrollers, along with a user-friendly waste management application. The application provides real-time updates, alerts, and data analytics to optimize waste collection and management processes. A real-time implementation model is also developed to demonstrate the system's efficiency and practicality. Overall, this smart waste management system enhances cleanliness, promotes environmental sustainability, and contributes to the development of smart cities by leveraging IoT and AI technologies.

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LIST OF ABBREVIATIONS

S.NO	ABBREVIATIONS	EXPANSIONS
1	RFID	Radio Frequency Identification
2	IOT	Internet of Things
3	WSN	Wireless Sensor Network
4	NTL	Non-Technical Loss
5	GPS	Global Positioning System
6	SWM	Solid Waste Management
7	HWM	Hazardous Waste Management

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

The rapid growth of population, urbanization, and industrialization has led to a significant increase in the generation of solid waste across the world. Managing this waste efficiently has become one of the most critical environmental challenges faced by modern society. In many developing countries, improper waste disposal and lack of segregation at the source contribute to severe environmental pollution, health hazards, and inefficient recycling processes. A major issue in the existing waste management system is the mixing of biodegradable (organic) and non-biodegradable (inorganic) waste, which makes recycling difficult and increases the burden on landfills. Traditional waste management systems rely heavily on manual labor for waste collection and segregation, which is time-consuming, error-prone, and often unhygienic. Additionally, the absence of real-time monitoring leads to problems such as overflowing bins, irregular waste collection, and inefficient route planning for waste collection vehicles.

These limitations highlight the need for an intelligent and automated system that can ensure proper waste segregation and efficient management.



Fig 1.1 Municipal solid waste testing service

With advancements in technologies such as the Internet of Things (IoT) and Artificial Intelligence (AI), it is now possible to develop smart and automated solutions for waste management. IoT enables the interconnection of physical devices, allowing real-time data collection and communication, while AI provides intelligent decision-making capabilities such as waste classification and predictive analysis. By integrating these technologies, waste

management systems can be transformed into more efficient, reliable, and scalable solutions. The proposed Smart Waste Management System aims to address these challenges by introducing smart bins equipped with sensors and embedded systems. These bins are capable of detecting the type of waste and segregating it into biodegradable and non-biodegradable categories automatically. Sensors such as ultrasonic sensors are used to measure the fill level of the bin, while moisture and gas sensors help in identifying the nature of waste and monitoring environmental conditions. The collected data is transmitted to a cloud platform through IoT communication modules, enabling continuous monitoring and control. Furthermore, the system incorporates a 24/7 monitoring mechanism that provides real-time updates on bin status, ensuring timely waste collection and preventing overflow conditions.

A user-friendly mobile or web application is also developed to display data, generate alerts, and assist authorities in managing waste collection efficiently. The integration of data analytics further helps in optimizing waste collection routes and improving overall system performance. In conclusion, this project presents a modern approach to waste management by leveraging IoT and AI technologies. It not only promotes proper waste segregation at the source but also enhances operational efficiency, reduces environmental impact, and contributes to the development of cleaner and smarter cities.

Waste management is the "generation, prevention, characterization, monitoring, treatment, handling, reuse and residual disposition of solid wastes". There are various types of solid waste including municipal (residential, institutional, commercial), agricultural, and special (health care, household hazardous wastes, sewage sludge).^[1] The term usually relates to materials produced by human activity, and the process is generally undertaken to reduce their effect on health, the environment or aesthetics.

There is a wide array of issues relating to waste management and those areas include:

- Generation of waste
- Waste minimization
- Waste removal
- Waste transportation
- Waste treatment
- Recycling and reuse
- Storage, collection, transport, and transfer

- Treatment
- Landfill disposal
- Environmental considerations
- Financial and marketing aspects
- Policy and regulations
- Education and training
- Planning and implementation.

Waste management practices are not uniform among: countries (developed and developing nations); regions (urban and rural area), and sectors (residential and industrial)

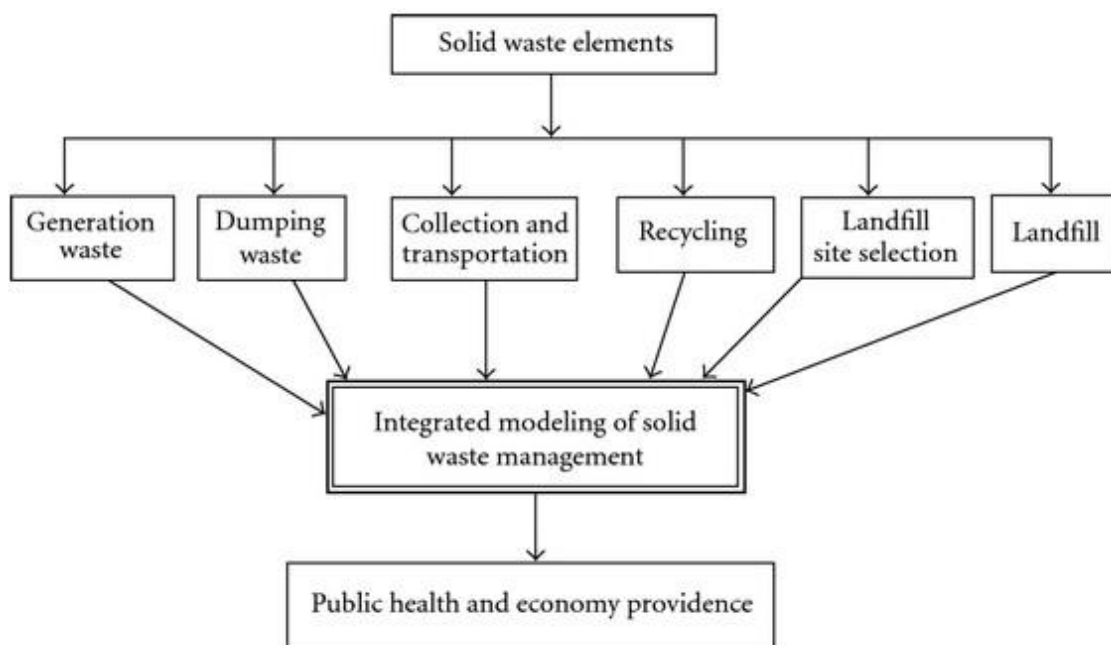
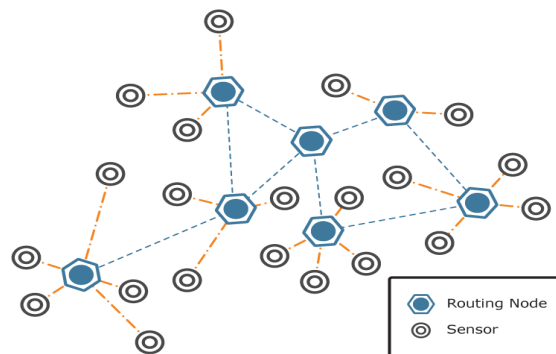


Fig 1.2 Integrated solid waste management

1.1 Wireless sensor network

A wireless sensor network is a group of specialized transducers with a communications infrastructure that uses radio to monitor and record physical or environmental conditions. A wireless sensor network is a group of specialized transducers with a communications infrastructure for monitoring and recording conditions at diverse locations. Commonly monitored parameters are temperature, humidity, pressure, wind direction and speed, illumination intensity, vibration intensity, sound intensity, power-line voltage, chemical concentrations, pollutant levels and vital body functions.

A sensor network consists of multiple detection stations called sensor nodes, each of which is small, lightweight and portable. Every sensor node is equipped with a transducer, microcomputer, transceiver and power source. The transducer generates electrical signals based on sensed physical effects and phenomena. The microcomputer processes and stores the sensor output. The transceiver receives commands from a central computer and transmits data to that computer. The power for each sensor node is derived from a battery.



1.2 Wireless sensor network architecture

Radio Frequency Identification (RFID) is one of the fastest growing wireless technologies this decade, mainly because it presents considerable advantages in terms of flexibility, lifetime and low cost. RFID offers great potential benefits in object identification and tracking in diverse application areas. RFID is above all an information and communication technology. Therefore, it is viewed as having a negative impact on the environment. Moreover, RFID tags are neither biodegradable nor recyclable. Nonetheless, RFID is also a wireless and powerless technology.

From the integration of sensor technology to tags, RFID presents new opportunities in many industrial applications. This paper presents more in details the following aspects: summary of RFID technology, presentation of green projects integrating RFID, expected progress and actual researches in RFID. During the last decades bar codes have probably been the most successful identification technology, as other procedures didn't yet exist or were too expensive to be deployed anciently, although technologies like RFID have many advantages. RFID technology is cheap, fast, does not require a line of sight and supports simultaneous reading of several tags at once over distances ranging from few centimeters to hundreds of meters. Due to technological progress the technology behind RFID has improved, lowering the costs to reasonable levels allowing it to enter the mass market.

Every RFID system consists of a reader, a transponder, and an application software. The transponder is a data-carrier which is commonly called tag. They come in various shapes and sizes, mostly in form of a so called smart cards or smart labels, which are sticky and can easily be attached to objects. Widespread applications can be found in the ends of anti-theft, access control, logistics, supply chain management, animal identification and tracking, and electronic payment systems, to name few examples for the countless uses of RFID.

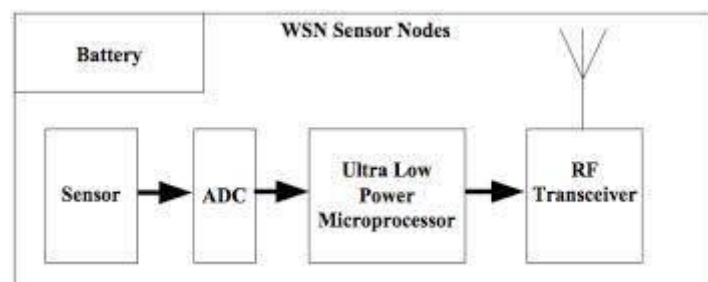


Fig 1.4 Block diagram of sensor

1.2 Applications

The applications for WSNs are many and varied. They are used in commercial and industrial applications to monitor data that would be difficult or expensive to monitor using wired sensors.

They could be deployed in wilderness areas, where they would remain for many years (monitoring some environmental variable) without the need to recharge/replace their power supplies. They could form a perimeter about a property and monitor the progression of intruders (passing information from one node to the next). There are a many uses for WSNs. Typical applications of WSNs include monitoring, tracking, and controlling. Some of the specific applications are habitat monitoring, object tracking, nuclear reactor controlling, fire detection, traffic monitoring, etc. In a typical application, a WSN is scattered in a region where it is meant to collect data through its sensor nodes.

- Environmental monitoring
- Habitat monitoring
- Acoustic detection
- Seismic Detection
- Military surveillance
- Inventory tracking
- Medical monitoring
- Smart spaces
- Process Monitoring

1.3 Area monitoring

Area monitoring is a typical application of WSNs. In area monitoring, the WSN is deployed over a region where some phenomenon is to be monitored. As an example, a large quantity of sensor nodes could be deployed over a battlefield to detect enemy intrusion instead of using landmines. When the sensors detect the event being monitored (heat, pressure, sound, light, electro-magnetic field, vibration, etc), the event needs to be reported to one of the base stations, which can take appropriate action (e.g., send a message on the internet or to a satellite). Depending on the exact application, different objective functions will require different data-propagation strategies, depending on things such as need for real-time response, redundancy of the data (which can be tackled via data aggregation techniques), need for security, etc.

1.4 Characteristics

Unique characteristics of a WSN are:

- Small-scale sensor nodes
- Limited power they can harvest or store
- Harsh environmental conditions
- Node failures
- Mobility of nodes
- Dynamic network topology
- Communication failures
- Heterogeneity of nodes
- Large scale of deployment
- Unattended operation

Sensor nodes can be imagined as small computers, extremely basic in terms of their interfaces and their components. They usually consist of a processing unit with limited computational power and limited memory, sensors (including specific conditioning circuitry), a communication device (usually radio transceivers or alternatively optical), and a power source usually in the form of a battery. Other possible inclusions are energy harvesting modules, secondary ASICs, and possibly secondary communication devices (e.g. RS232 or USB). The base stations are one or more distinguished components of the WSN with much more computational, energy and communication resources. They act as a gateway between sensor nodes and the end user.

Generation, transmission and distribution of electrical energy involve many operational losses. Whereas, losses implicated in generation can be technically defined, but T&D losses cannot be precisely quantified with the sending end information. This illustrates the involvement of non-technical parameters in T&D of electricity. Overall technical losses occur naturally and are caused because of power dissipation in transmission lines, transformers, and other power system components. Technical losses in T&D are computed with the information about total load and the total energy billed. NTL cannot be precisely computed, but can be estimated from the

difference between the total energy supplied to the customers and the total energy billed. NTL are caused by the factors external to the power system. In many developing countries, NTL are a serious concern for utility companies as they account to about 10 to 40% of their total generation capacity. Data regarding NTL is uncertain and it is very difficult to analyze theft in terms of actions that cause these losses. Electricity theft forms a major chunk of the NTL. Electricity theft includes bypassing, tampering with the energy meter and other physical methods to evade payment. Illegal tapping of electricity from the feeder and tampering with the meter are the most identified and accounted ways of theft.

Electricity theft can also be defined as, using electricity from the utility company without a contract or valid obligation to alter its measurement is called electricity theft. Electricity theft is a complex problem with many parameters to evaluate. These parameters include political, economical, criminal, and managerial issues. Apart from illustrated methods, billing irregularities is another form of NTL in many developing countries. Corruption and lack of commitment in the utility companies to control the illegal consumption are the reasons for such irregularities. For instance, an employee might not read the correct value of the total energy displayed by energy meter or may intentionally give lower value than the exact one.

Few household and industrial customers are responsible for nonpayment of bills to save their business interests in the cover of weak economic situations.. As the extent of these losses is very large, it is essentially important to force the utility companies to design mechanism that reduce NTL.. In light of this issue, India incurs losses of about \$9 billion every year in the form of electricity theft. In many countries electric power theft is annually sized up in billions dollars. To lower volumes of theft of the electric power, power stations of the general use optimize the data gained by a new Developed Infrastructure of Measurements (RII), and use their analysis for establishment of indexes of consumption deviating norm and possible swindle.

In the given research we offer prime strategy of usage of the analysis of convergence for establishment of cases of larceny of the electric power and new principles of measurement which help to optimize these strategy for an estimation and comparing of the anomalous indexes. We use real-life convergence of system

As intelligent network is called upgrade of an infrastructure of a power supply system by means of new production engineering that allows automatic system to show the big efficiency, reliability and safety, providing at the same time the big transparency and possibility of a choice for customers. One of the key production engineering actively involved worldwide is production engineering of the Developed Infrastructure of Measurements(RII).

RII is the upgraded system of measurement where old mechanical instruments are replaced new by counters. Such instruments are again gated in devices providing a two-way communication between the energetic companies and customers, in a consequence of that necessity to send the employee of service for removal of indexes disappears, and there are new possibilities, such as possibility to track current consumption volume in more details, to detect failures with power company analog instruments learned about failures in the core through complaints of customers automatically more sweepingly to recover power supply, to carry out remote uncoupling and to send to customers the information (for example, about a time history of the prices or restarted sources of the electric power) to give to customers more than the accessible information on their power consumption.

Counters represent billions devices of wide consumption with an operation life in some decades and acting in physically unsafe places. Complication of these devices by means of adding of processors and secure memory can increase the price of clever counters for some dollars and as the energetic companies are forced to involve million devices, the market reality is that that in practice they appear uneconomical and are not recommended in the capacity of necessary.

CHAPTER 2

LITERATURE SURVEY

2.1 Electronic Waste and the Special Case of Lead-Free Piezoelectrics, H. Alex Hsain, 2019, IEEE

The integration of communication technologies such as radio frequency identification (RFID), global positioning system (GPS), general packet radio system (GPRS), geographic information system (GIS) with a camera are constructed for solid waste monitoring system. The aim is to improve the way of responding to customer's inquiry and emergency cases and estimate the solid waste amount without any involvement of the truck driver. The proposed system consists of RFID tag mounted on the bin, RFID reader as truck module, GPRS/GSM as web-server, GIS as map server, database server and control station server.

The tracking devices mounted in the trucks collect location information in real-time via the GPS. This information is transferred continuously through GPRS to a central database. The users are able to view the current location of each truck in the collection stage via a web-based application, and thereby manage the fleet. The trucks positions and trash bin information are displayed on a digital map, which is made available by a map server. Thus, the solid waste of the bin and the truck are being monitored using the developed system.

2.2 Global and Indian Perspective of E-Waste and its Environmental Impact, Vinay Kumar Pant¹ and Sumit Kumar²,2018,IEEE

Radio frequency identification (RFID) is one of the most promising and anticipated technologies in recent years. This paper explores the potential of RFID in efficient solid waste management. With increasing urbanisation, the need to manage solid wastes in an environment-friendly way is a daunting task for any city. This paper proposes an automated system for waste collection and container monitoring system using RFID, global positioning systems (GPS), geographical information systems (GIS) and GSM (Global System for Mobile Communications). RFID is a generic term for technologies that use radio waves to identify people, animals or objects without being in physical or visual contact with them.

The proposed system consists of RFID tags mounted on containers, RFID readers mounted on trucks along with GPS for location tracking and GSM module for wireless transmission. The system provides real time monitoring of waste collection through a web based application. This application helps administrators in decision making related to reallocation of routes and containers, etc. and handling management issues such as observing performance of contractors and waste generation characteristics of a particular area. It also enhances transparency in the working of the civic administration.

2.3 Mechanism of Metal Removal from Metal-Coated Plastics Using Pulsed Power, Hamid Hosano, Hidenori Akiyama and Takashi Sakugawa,2019,IEEE

Solid Waste Management (SWM) is the generation, separation, collection, transfer, transportation and disposal of waste in a way that takes into account public health, economics, conservation, aesthetics, and the environment, and is responsive to public demands. Failure of the municipal solid waste management system has also serious environmental impacts like land and air pollution, blockage of drains and water pollution in natural streams. The objective of this review was to carry out the situational analysis of solid waste management in South Asian countries particularly in the context of Pakistan. A review of the literature was done using Google scholar, Pubmed and Pakmedinet as search engines.

Books were also consulted that provides information about SWM. There are various factors that attribute to poor solid waste management, such as, lack of public awareness, unplanned city growth, high waste generation and non-functioning of existing systems. Rate of urbanization, scavenger role for recyclable separation and the capacities of existing municipalities for solid waste management are also important factors that should be considered. This review concluded that unfortunately solid waste is not considered a big problem in Pakistan and health authorities fail to appreciate the magnitude of the problem. Participation of different sectors including community ownership, role of government and private sector is important for better management of solid waste. Regulatory authorities have to make certain rules and most importantly find ways to implement it.

2.4 Does the Entry of Third-Party Remanufacturers Always Hurt Original Equipment Manufacturers, Xiaole Wu , Yu Zhou, 2015.Ddecision Sciences

Hazardous Waste Management (HWM) is of global significance. The adverse impacts caused due to the indiscriminate disposal of Hazardous Wastes (HWs) are considered as technological disasters. There was no proper secured landfill facility available in India to dispose of Hazardous Waste (HW) till 1997. Very few industries in India, mostly in large scale and a few in medium scale, own proper treatment and disposal facilities. A common waste treatment and disposal facility such as Treatment, Storage and Disposal Facility (TSDF) for management of HWs generated from industries is one of the useful options under such conditions. Few Guidelines issued by Ministry of Environment and Forests under Hazardous Wastes (Management & Handling) Rules, 1989 promulgated under Environment (Protection) Act, 1986 are available in India for selection of best site for TSDF.

The planning for HWM comprises of several aspects ranging from identification and quantification of HW to development and monitoring of TSDF. This paper focuses on the basic steps involved in the Comprehensive HWM. The physical models developed by the authors for ranking of TSDF sites based on the Guidelines available are discussed. The current status in India pertaining to generation of HW and the TSDF sites is also addressed.

2.5 Industrial Marketing Management, Thomas E. Johnsen, Joe Miemczyk ,2016, Elsevier

This paper gives an overview of the current state of radio frequency identification (RFID) technology. Aside from a brief introduction to the principles of the technology, major current and envisaged fields of application, as well as advantages, and limitations of use are discussed. Radio frequency identification (RFID) is a generic term that is used to describe a system that transmits the identity (in the form of a unique serial number) of an object or person wirelessly, using radio waves. It's grouped under the broad category of automatic identification technologies. Typically, recording the collection process and estimating the area condition are a manual operation which is the responsibility of the truck driver.

Although some of solid waste companies have involved GPS system to capture the time spent but they cannot get the accurate bin information. However, manual operations can be misread, misreported, mistyped or left blank altogether. Furthermore, asking drivers to manually collect the data is a significant distraction to their main responsibility of driving. Figure 1 shows the tag attached with bin and RFID reader and camera associated with on-board GPS and GSM module are attached with truck for solid waste bin monitoring system. FRID is increasingly used with biometric technologies for security. In this paper Basic Principles of RFID technology along with its types are discussed.

CHAPTER 3

EXISTING SYSTEM

The current waste management system in most regions primarily relies on traditional and manual methods of waste collection and disposal. Waste generated from households, commercial areas, and public places is typically collected in common bins without proper segregation into biodegradable and non-biodegradable categories. This lack of segregation at the source creates significant challenges in recycling, composting, and safe disposal processes. In the existing system, waste bins are usually monitored manually by sanitation workers. There is no real-time mechanism to track the fill level or condition of the bins. As a result, bins often overflow before being emptied, leading to unhygienic conditions, foul odors, and environmental pollution. Overflowing waste also attracts pests and contributes to the spread of diseases, posing serious health risks to the public.

Another limitation of the traditional system is the inefficiency in waste collection. Collection vehicles follow fixed schedules and routes, regardless of whether the bins are full or empty. This results in unnecessary fuel consumption, increased operational costs, and inefficient use of manpower. At the same time, some bins may remain unattended for long periods, leading to accumulation of waste. The process of waste segregation, if performed, is usually done manually at dumping yards or processing units. This method is labor-intensive, time-consuming, and often inaccurate. Manual segregation also exposes workers to hazardous materials and harmful gases, affecting their health and safety. Furthermore, improper handling of mixed waste reduces the quality of recyclable materials and limits the effectiveness of waste processing techniques.

Additionally, the existing system lacks proper data collection and analysis. There is no centralized platform to monitor waste generation patterns, bin status, or collection efficiency. Without data-driven insights, it becomes difficult for authorities to make informed decisions or improve the overall waste management process. Overall, the traditional waste management system is inefficient, unorganized, and unable to meet the growing demands of modern urban environments. These limitations highlight the need for a smart, automated, and technology-driven solution that ensures proper waste segregation, real-time monitoring, and efficient waste collection.

CHAPTER 4

PROPOSED SYSTEM

To overcome the limitations of the traditional waste management system, this project proposes a Smart Waste Management System that utilizes Internet of Things (IoT) and Artificial Intelligence (AI) technologies for efficient waste segregation, monitoring, and management. The system is designed to promote proper disposal of waste at the source level and ensure real-time tracking of waste bins. The core component of the proposed system is a smart bin integrated with various sensors and a microcontroller (such as Arduino or ESP32). The smart bin is capable of automatically identifying and segregating waste into biodegradable and non-biodegradable categories. Sensors such as moisture sensors are used to detect wet (organic) waste, while additional sensors like metal detectors or infrared sensors can assist in identifying dry waste materials.

In advanced implementations, AI-based image processing techniques can be used to improve the accuracy of waste classification. Once the waste is identified, the system uses a mechanical arrangement (such as servo motors or flaps) to direct the waste into the appropriate compartment within the bin. This automation minimizes human intervention and ensures accurate segregation at the source itself. The system also includes an ultrasonic sensor to continuously monitor the fill level of the bin. When the waste level reaches a predefined threshold, the system automatically sends alerts to the concerned authorities or waste collection personnel. This prevents overflow and maintains cleanliness in the surrounding environment. For communication, the system uses IoT modules such as Wi-Fi (ESP8266/ESP32) or GSM to transmit real-time data to a cloud platform. The data includes bin status, waste level, and environmental parameters such as gas concentration. This enables a 24/7 monitoring system where authorities can track multiple bins remotely. A mobile or web-based application is developed as part of the system to provide a user-friendly interface.

4.1 BLOCK DIAGRAM

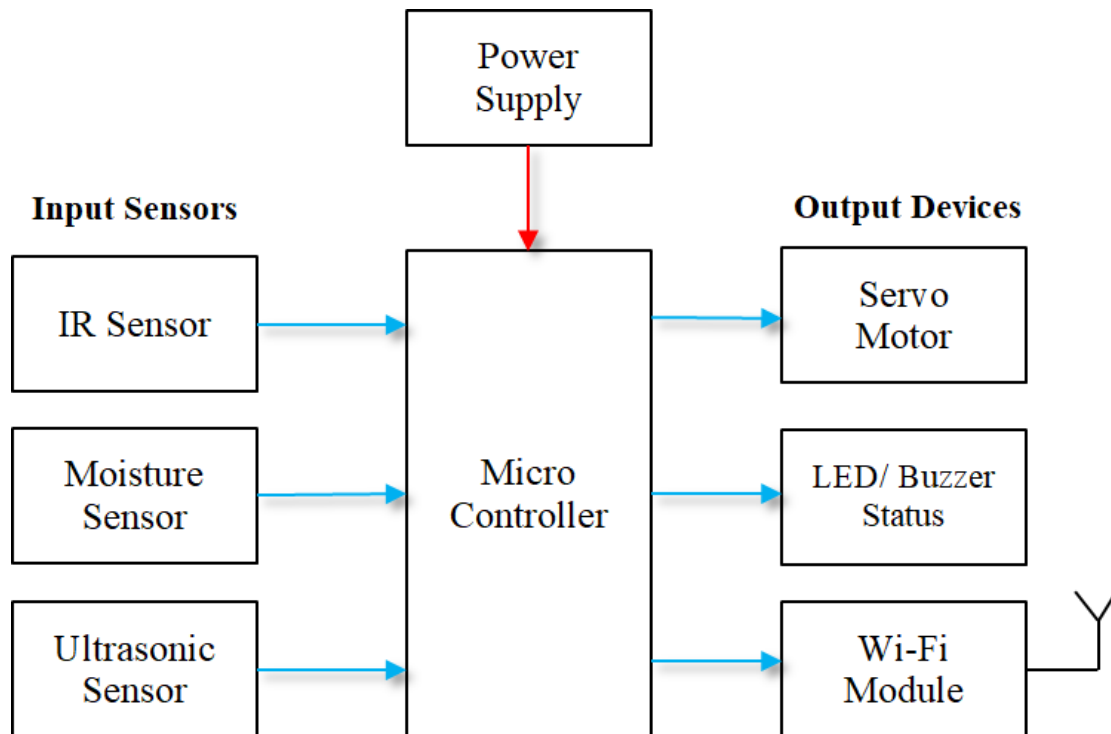


Fig 4.1 IOT Based monitoring system

CHAPTER 5

HARDWARE DESCRIPTION

5.1 ESP 32

The ESP32 *Microcontroller Unit* is an open-source software and hardware development environment built around an inexpensive System-on-a-Chip (SoC) called the ESP32. The ESP32, designed and manufactured by Espressif Systems, contains the crucial elements of a computer: CPU, RAM, networking (WiFi), and even a modern operating system and SDK. That makes it an excellent choice for the Internet of Things (IoT) projects of all kinds. However, as a chip, the ESP32 is also hard to access and use. You must solder wires, with the appropriate analog voltage, to its pins for the simplest tasks such as powering it on or sending a keystroke to the “computer” on the chip. You also have to program it in low-level machine instructions that can be interpreted by the chip hardware. This level of integration is not a problem using the ESP32 as an embedded controller chip in mass-produced electronics. It is a huge burden for hobbyists, hackers, or students who want to experiment with it in their own IoT projects.

But, what about Arduino? The Arduino project created an open-source hardware design and software SDK for their versatile IoT controller. Similar to NodeMCU, the Arduino hardware is a microcontroller board with a USB connector, LED lights, and standard data pins. It also defines standard interfaces to interact with sensors or other boards. But unlike NodeMCU, the Arduino board can have different types of CPU chips (typically an ARM or Intel x86 chip) with memory chips, and a variety of programming environments. There is an Arduino reference design for the ESP32 chip as well. However, the flexibility of Arduino also means significant variations across different vendors. For example, most Arduino boards do not have WiFi capabilities, and some even have a serial data port instead of a USB port

5.1.1 USB to Serial Converter - CP2102 or CH340G

Incorporated into each NodeMCU is a USB to Serial Converter. The official design is based on the CP2102 chipset and offers the best compatibility. Genuine boards use the CP2102 chipset including the officially licensed Amica NodeMCU modules. The other common USB to Serial Converter used is the CH340G which is common on the lower-priced modules including the LoLin units. Other designs may use drivers including the FTDI chipset, but those designs are rare.

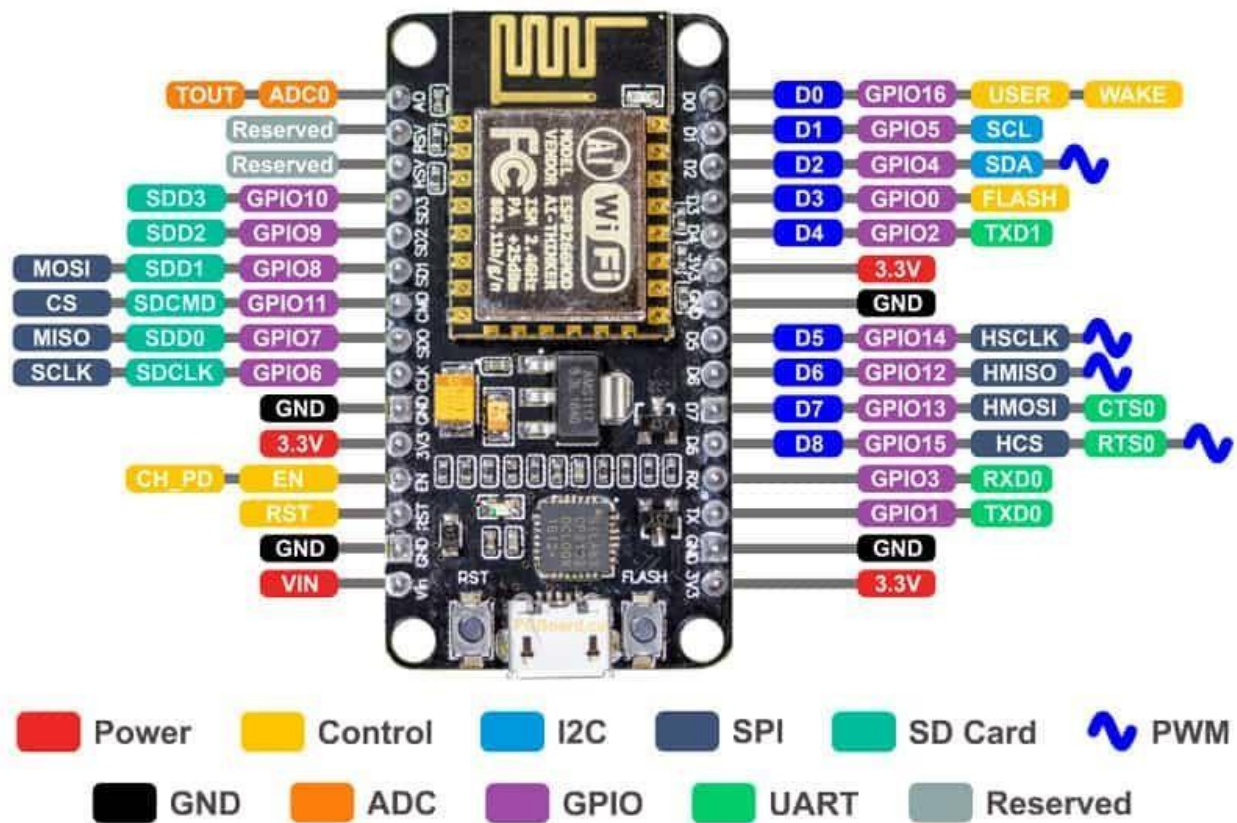


Fig 5.1 ESP8266 Node MCU

- Power pin - There are four power pins. VIN pin and three 3.3V pins.
- VIN can be used to directly supply the NodeMCU/ESP32 and its peripherals. Power delivered on VIN is regulated through the onboard regulator on the NodeMCU module – you can also supply 5V regulated to the VIN pin

- **3.3V** pins are the output of the onboard voltage regulator and can be used to supply power to external components.
- GND are the ground pins of NodeMCU/ESP32
- I2C PINS are used to connect I2C sensors and peripherals. Both I2C Master and I2C Slave are supported. I2C interface functionality can be realized programmatically, and the clock frequency is 100 kHz at a maximum. It should be noted that I2C clock frequency should be higher than the slowest clock frequency of the slave device.
- GPIO Pins - NodeMCU/ESP32 has 17 GPIO pins which can be assigned to functions such as I2C, I2S, UART, PWM, IR Remote Control, LED Light and Button programmatically. Each digital enabled GPIO can be configured to internal pull-up or pull-down, or set to high impedance. When configured as an input, it can also be set to edge-trigger or level-trigger to generate CPU interrupts.
- ADC Channel - The NodeMCU is embedded with a 10-bit precision SAR ADC. The two functions can be implemented using ADC. Testing power supply voltage of VDD3P3 pin and testing input voltage of TOUT pin. However, they cannot be implemented at the same time.
- UART Pins - NodeMCU/ESP32 has 2 UART interfaces (UART0 and UART1) which provide asynchronous communication (RS232 and RS485), and can communicate at up to 4.5 Mbps. UART0 (TXD0, RXD0, RST0 & CTS0 pins) can be used for communication. However, UART1 (TXD1 pin) features only data transmit signal so, it is usually used for printing log.

- SPI Pins - NodeMCU/ESP32 features two SPIs (SPI and HSPI) in slave and master modes. These SPIs also support the following general-purpose SPI features:
 - 4 timing modes of the SPI format transfer
 - Up to 80 MHz and the divided clocks of 80 MHz
 - Up to 64-Byte FIFO
- SDIO Pins - NodeMCU/ESP32 features Secure Digital Input/Output Interface (SDIO) which is used to directly interface SD cards. 4-bit 25 MHz SDIO v1.1 and 4-bit 50 MHz SDIO v2.0 are supported.
- PWM Pins - The board has 4 channels of Pulse Width Modulation (PWM). The PWM output can be implemented programmatically and used for driving digital motors and LEDs. PWM frequency range is adjustable from 1000 μ s to 10000 μ s (100 Hz and 1 kHz).
- Control Pins are used to control the NodeMCU/ESP32. These pins include Chip Enable pin (EN), Reset pin (RST) and WAKE pin.
 - **EN:** The ESP32 chip is enabled when EN pin is pulled HIGH. When pulled LOW the chip works at minimum power.
 - **RST:** RST pin is used to reset the ESP32 chip.
 - **WAKE:** Wake pin is used to wake the chip from deep-sleep.
- Control Pins are used to control the NodeMCU/ESP32. These pins include Chip Enable pin (EN), Reset pin (RST) and WAKE pin.
 - **EN:** The ESP32 chip is enabled when EN pin is pulled HIGH. When pulled LOW the chip works at minimum power.
 - **RST:** RST pin is used to reset the ESP32 chip.
 - **WAKE:** Wake pin is used to wake the chip from deep-sleep.

Depending on the Operating System you are using with the NodeMCU, the appropriate driver must be installed. Generally, Windows 10 immediately recognizes the CP2102 chipset while the CH340G may require separate installation.

Drivers for the CP2102 are available for **download** from the Silicon Labs support site. Drivers constantly evolve and ensuring and installing the most recent version in your development environment minimum issues. Drivers are available for Windows, Mac, Linux, and Android. *We also have a local copy of the CP2102 drivers (v10.1.8) available locally for download. You always best to visit the original manufacturer to ensure you are receiving the most recent versions of the driver.*



Fig 5.2 Regulated DC Power supply

WCH maintain and update the drivers for the CH340G on a regular basis. Versions of the driver are also available for Windows, Mac, Linux, and Android. Visit their Driver Download page. We also have a local copy of the CH340G drivers (version 3.5) available locally for download. *You are always best to visit the original manufacturer to ensure you are receiving the most recent versions of the driver*

5.2 POWER SUPPLY

5.2.1 Block diagram

The ac voltage, typically 220V rms, is connected to a transformer, which steps that ac voltage down to the level of the desired dc output. A diode rectifier then provides a full-wave rectified voltage that is initially filtered by a simple capacitor filter to produce a dc voltage. This resulting dc voltage usually has some ripple or ac voltage variation.

A regulator circuit removes the ripples and also remains the same dc value even if the input dc voltage varies, or the load connected to the output dc voltage changes. This voltage regulation is usually obtained using one of the popular voltage regulator IC units.



5.2.2 Working principle

5.2.2.1 Transformer

The potential transformer will step down the power supply voltage (0-230V) to (0-6V) level. Then the secondary of the potential transformer will be connected to the precision rectifier, which is constructed with the help of op-amp. The advantages of using precision rectifier are it will give peak voltage output as DC; rest of the circuits will give only RMS output.

5.2.2.2 Bridge rectifier

When four diodes are connected as shown in figure, the circuit is called as bridge rectifier. The input to the circuit is applied to the diagonally opposite corners of the network, and the output is taken from the remaining two corners. Let us assume that the transformer is working properly and there is a positive potential, at point A and a negative potential at point B. the positive potential at point A will forward bias D3 and reverse bias D4.

The negative potential at point B will forward bias D1 and reverse D2. At this time D3 and D1 are forward biased and will allow current flow to pass through them; D4 and D2 are reverse biased and will block current flow. The path for current flow is from point B through D1, up through RL, through D3, through the secondary of the transformer back to point B. this path is indicated by the solid arrows. Waveforms (1) and (2) can be observed across D1 and D3.

One-half cycle later the polarity across the secondary of the transformer reverse, forward biasing D2 and D4 and reverse biasing D1 and D3. Current flow will now be from point A through D4, up through RL, through D2, through the secondary of T1, and back to point A. This path is indicated by the broken arrows. Waveforms (3) and (4) can be observed across D2 and D4. The current flow through RL is always in the same direction. In flowing through RL this current develops a voltage corresponding to that shown waveform (5). Since current flows through the load (RL) during both half cycles of the applied voltage, this bridge rectifier is a full-wave rectifier.

One advantage of a bridge rectifier over a conventional full-wave rectifier is that with a given transformer the bridge rectifier produces a voltage output that is nearly twice that of the conventional full-wave circuit. This may be shown by assigning values to some of the components shown in views A and B. assume that the same transformer is used in both circuits. The peak voltage developed between points X and y is 1000 volts in both circuits. In the conventional full-wave circuit shown—in view A, the peak voltage from the center tap to either X or Y is 500 volts. Since only one diode can conduct at any instant, the maximum voltage that can be rectified at any instant is 500 volts.

The maximum voltage that appears across the load resistor is nearly-but never exceeds-500 volts, as result of the small voltage drop across the diode. In the bridge rectifier shown in view B, the maximum voltage that can be rectified is the full secondary voltage, which is 1000 volts. Therefore, the peak output voltage across the load resistor is nearly 1000 volts. With both circuits using the same transformer, the bridge rectifier circuit produces a higher output voltage than the conventional full-wave rectifier circuit.

5.2.2.3 IC Voltage regulators

Voltage regulators comprise a class of widely used ICs. Regulator IC units contain the circuitry for reference source, comparator amplifier, control device, and overload protection all in a single IC. IC units provide regulation of either a fixed positive voltage, a fixed negative voltage, or an adjustably set voltage. The regulators can be selected for operation with load currents from hundreds of milli amperes to tens of amperes, corresponding to power ratings from milli watts to tens of watts.

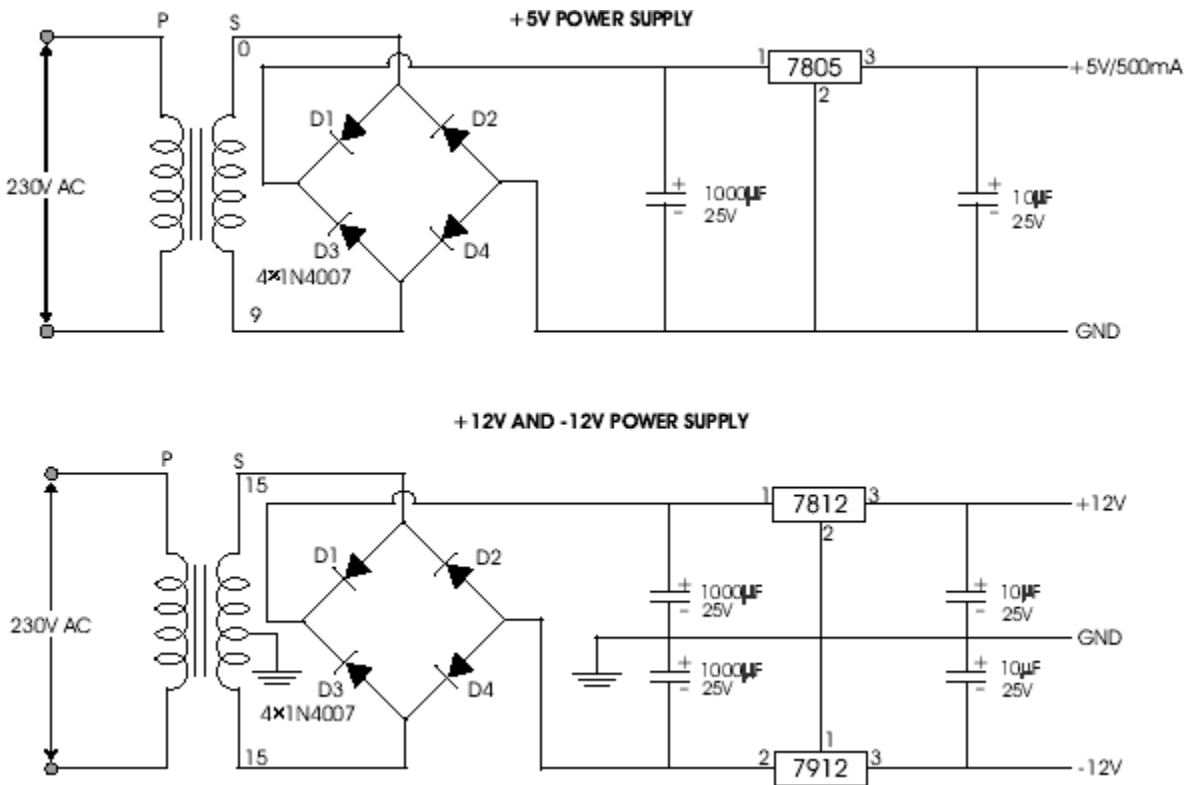


Fig 5.3 IC Voltage regulator

A fixed three-terminal voltage regulator has an unregulated dc input voltage, V_i , applied to one input terminal, a regulated dc output voltage, V_o , from a second terminal, with the third terminal connected to ground. The series 78 regulators provide fixed positive regulated voltages from 5 to 24 volts. Similarly, the series 79 regulators provide fixed negative regulated voltages from 5 to 24 volts.

- For ICs, microcontroller, LCD 5 volts
- For alarm circuit, op-amp, relay circuits 12 volts

5.3 INFRARED SENSOR

An infrared sensor is an electronic device that emits and/or detects infrared radiation in order to sense some aspect of its surroundings. Infrared sensors can measure the heat of an object, as well as detect motion. Many of these types of sensors only measure infrared radiation, rather than emitting it, and thus are known as passive infrared (PIR) sensors.

All objects emit some form of thermal radiation, usually in the infrared spectrum. This radiation is invisible to our eyes, but can be detected by an infrared sensor that accepts and interprets it. In a typical infrared sensor like a motion detector, radiation enters the front and reaches the sensor itself at the center of the device. This part may be composed of more than one individual sensor, each of them being made from pyroelectric materials, whether natural or artificial. These are materials that generate an electrical voltage when heated or cooled.

These pyroelectric materials are integrated into a small circuit board. They are wired in such a way so that when the sensor detects an increase in the heat of a small part of its field of view, it will trigger the motion detector's alarm. It is very common for an infrared sensor to be integrated into motion detectors like those used as part of a residential or commercial security system.



Fig 5.4 IR Sensor

5.3.1 Emissivity

- Emissivity is the measurement of energy-emitting characteristics of different materials or surfaces. An object's emitted, or produced, IR energy decreases or increases in proportion to its temperature. IR sensors use adjustable emissivity settings that enable them to obtain temperature measurements of different surface types.

5.3.2 Detection

- An object's emitted energy reaches the IR sensor via the device's optic system, which in turn focuses the energy onto its photosensitive detectors. The detectors convert the IR energy into an electrical signal, which then is converted into a temperature value based on the object's emissivity. This value may be displayed on the IR sensor or converted as digital input and sent to a computer terminal for display.

5.4 Ultrasonic sensor

Ultrasonic sensors (also known as **transceivers** when they both send and receive) work on a principle similar to radar or sonar **which** evaluate attributes of a target by interpreting the echoes from radio or sound waves respectively. Ultrasonic sensors generate high frequency sound waves and evaluate the echo which is received back by the sensor. Sensors calculate the time interval between sending the signal and receiving the echo to determine the distance to an object.

This technology can be used for measuring: wind speed and direction (anemometer), fullness of a tank and speed through air or water. For measuring speed or direction a device uses multiple detectors and calculates the speed from the relative distances to particulates in the air or water. To measure the amount of liquid in a tank, the sensor measures the distance to the surface of the fluid. Further applications include: humidifiers, sonar, medical ultrasonography, burglar alarms and non-destructive testing.

Systems typically use a transducer which generates sound waves in the ultrasonic range, above 18,000 hertz, by turning electrical energy into sound, then upon receiving the echo turn the sound waves into electrical energy which can be measured and displayed.

The technology is limited by the shapes of surfaces and the density or consistency of the material. For example foam on the surface of a fluid in a tank could distort a reading.



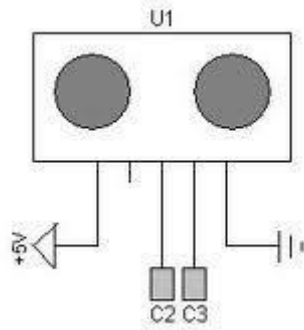
Fig 5.5 Ultrasonic sensor recognition

5.4.1 Transducers

An ultrasonic transducer is a device that converts energy into ultrasound, or sound waves above the normal range of human hearing. While technically a dog whistle is an ultrasonic transducer that converts mechanical energy in the form of air pressure into ultrasonic sound waves, the term is more apt to be used to refer to piezoelectric transducers that convert electrical energy into sound. Piezoelectric crystals have the property of changing size when a voltage is applied, thus applying an alternating current (AC) across them causes them to oscillate at very high frequencies, thus producing very high frequency sound waves.

Ultrasonic sensor provides a very low-cost and easy method of distance measurement. This sensor is perfect for any number of applications that require you to perform measurements between moving or stationary objects. Naturally, robotics applications are very popular but you'll also find this product to be useful in security systems or as an infrared replacement if so desired. You will definitely appreciate the activity status LED and the economic use of just one I/O pin.

5.4.2 Circuit diagram



Features:

- Provides precise, non-contact distance measurements within a 2 cm to 3 m range
- Simple pulse in/pulse out communication
- Burst indicator LED shows measurement in progress
- 20 mA power consumption
- Narrow acceptance angle
- 3-pin header makes it easy to connect using a servo extension cable, no soldering required

5.5 SERVO MOTOR

A **servo motor** is a specialized rotary actuator used for precise control of angular position, speed, and acceleration. In your automated toll plaza system, the servo motor plays a crucial role in controlling the opening and closing of the toll gate automatically based on the verification result received from the NodeMCU. A typical servo motor consists of three main components: a DC motor, a control circuit, and a feedback mechanism (usually a potentiometer). Unlike normal motors that rotate continuously, a servo motor rotates to a specific angle based on the control signal it receives. This makes it highly suitable for applications where accurate positioning is required, such as robotic arms, automation systems, and gate control mechanisms. The servo

motor operates using Pulse Width Modulation (PWM) signals. The NodeMCU sends PWM signals to the servo motor, where the width of the pulse determines the position of the motor shaft. For example, a pulse width of around 1 ms may correspond to 0°, 1.5 ms to 90°, and 2 ms to 180°. Based on these signals, the servo motor rotates to the desired angle and holds that position until a new signal is received.

In the proposed toll plaza system, when a vehicle is authenticated successfully, the NodeMCU sends a signal to the servo motor to rotate (for example, from 0° to 90°), which opens the toll gate. After a predefined delay, the motor returns to its original position (90° to 0°), thereby closing the gate. This ensures smooth and automatic operation without manual intervention. Servo motors are preferred in this project due to their high accuracy, quick response, low power consumption, and ease of interfacing with microcontrollers like NodeMCU. They are also compact, reliable, and capable of holding a fixed position without continuous input, making them ideal for automated gate control systems. Overall, the servo motor is an essential component in this system, enabling precise and efficient control of the toll gate, contributing to automation, reliability, and improved system performance.

CHAPTER 6

SYSTEM REQUIREMENTS

6.1 HARDWARE REQUIREMENTS:

CPU Processor	:	Intel Pentium
Hard Disk	:	40 GB
RAM	:	256 MB
Monitor	:	15VGA Color
Mouse	:	Ball/Optical
Keyboard	:	102 Keys

6.2 SOFTWARE REQUIREMENTS:

Operating System	:	Windows 8
Front End	:	PHP,Proteus
Back End	:	Embedded C ,My-SQL

Embedded C

An embedded system is an application that contains at least one programmable computer (typically in the form of a microcontroller, a microprocessor or digital signal processor chip) and which is used by individuals who are, in the main, unaware that the system is computer-based.

Introduction

Looking around, we find ourselves to be surrounded by various types of embedded systems. Be it a digital camera or a mobile phone or a washing machine, all of them has some kind of processor functioning inside it. Associated with each processor is the embedded software. If hardware forms the body of an embedded system, embedded processor acts as the brain, and embedded software forms its soul. It is the embedded software which primarily governs the functioning of embedded systems.

During infancy years of microprocessor based systems, programs were developed using assemblers and fused into the EPROMs. There used to be no mechanism to find what the program was doing. LEDs, switches, etc. were used to check correct execution of the program. Some ‘very fortunate’ developers had In-circuit Simulators (ICEs), but they were too costly and were not quite reliable as well.

As time progressed, use of microprocessor-specific assembly-only as the programming language reduced and embedded systems moved onto C as the embedded programming language of choice. C is the most widely used programming language for embedded processors/controllers. Assembly is also used but mainly to implement those portions of the code where very high timing accuracy, code size efficiency, etc. are prime requirements.

Initially C was developed by Kernighan and Ritchie to fit into the space of 8K and to write (portable) operating systems. Originally it was implemented on UNIX operating systems. As it was intended for operating systems development, it can manipulate memory addresses. Also, it allowed programmers to write very compact codes. This has given it the reputation as the language of choice for hackers too.

As assembly language programs are specific to a processor, assembly language didn't offer portability across systems. To overcome this disadvantage, several high level languages, including C, came up. Some other languages like PLM, Modula-2, Pascal, etc. also came but couldn't find wide acceptance. Amongst those, C got wide acceptance for not only embedded systems, but also for desktop applications. Even though C might have lost its sheen as mainstream language for general purpose applications, it still is having a strong-hold in embedded programming. Due to the wide acceptance of C in the embedded systems, various kinds of support tools like compilers & cross-compilers, ICE, etc. came up and all this facilitated development of embedded systems using C. Subsequent sections will discuss what is Embedded C, features of C language, similarities and difference between C and embedded C, and features of embedded C programming.

6.3 EMBEDDED SYSTEMS PROGRAMMING

Embedded systems programming is different from developing applications on a desktop computers. Key characteristics of an embedded system, when compared to PCs, are as follows. Embedded devices have resource constraints (limited ROM, limited RAM, limited stack space, less processing power). Components used in embedded system and PCs are different; embedded systems typically uses smaller, less power consuming components. Embedded systems are more tied to the hardware. Two salient features of Embedded Programming are code speed and code size. Code speed is governed by the processing power, timing constraints, whereas code size is

governed by available program memory and use of programming language. Goal of embedded system programming is to get maximum features in minimum space and minimum time.

6.3.1 Embedded systems are programmed using different type of language

- Machine Code
- Low level language, i.e., assembly
- High level language like C, C++, Java, Ada, etc.
- Application level language like Visual Basic, scripts, Access, etc.

Assembly language maps mnemonic words with the binary machine codes that the processor uses to code the instructions. Assembly language seems to be an obvious choice for programming embedded devices. However, use of assembly language is restricted to developing efficient codes in terms of size and speed. Also, assembly codes lead to higher software development costs and code portability is not there. Developing small codes are not much of a problem, but large programs/projects become increasingly difficult to manage in assembly language. Finding good assembly programmers has also become difficult nowadays. Hence high level languages are preferred for embedded systems programming.

Use of C in embedded systems is driven by following advantages it is small and reasonably simpler to learn, understand, program and debug. C Compilers are available for almost all embedded devices in use today, and there is a large pool of experienced C programmers.

Unlike assembly, C has advantage of processor-independence and is not specific to any particular microprocessor/ microcontroller or any system. This makes it convenient for a user to develop programs that can run on most of the systems. As C combines functionality of assembly language and features of high level languages, C is treated as a ‘middle-level computer language’

or ‘high level assembly language’. It is fairly efficient. It supports access to I/O and provides ease of management of large embedded projects.

Many of these advantages are offered by other languages also, but what sets C apart from others like Pascal, FORTRAN, etc. is the fact that it is a middle level language; it provides direct hardware control without sacrificing benefits of high level languages. Compared to other high level languages, C offers more flexibility because C is relatively small, structured language; it supports low-level bit-wise data manipulation.

Compared to assembly language, C Code written is more reliable and scalable, more portable between different platforms (with some changes). Moreover, programs developed in C are much easier to understand, maintain and debug. Also, as they can be developed more quickly, codes written in C offers better productivity. C is based on the philosophy ‘programmers know what they are doing’; only the intentions are to be stated explicitly. It is easier to write good code in C & convert it to an efficient assembly code (using high quality compilers) rather than writing an efficient code in assembly itself. Benefits of assembly language programming over C are negligible when we compare the ease with which C programs are developed by programmers.

Object oriented language, C++ is not apt for developing efficient programs in resource constrained environments like embedded devices. Virtual functions & exception handling of C++ are some specific features that are not efficient in terms of space and speed in embedded systems. Sometimes C++ is used only with very few features, very much as C.

Ada, also an object-oriented language, is different than C++. Originally designed by the U.S. DOD, it didn’t gain popularity despite being accepted as an international standard twice (Ada83 and Ada95). However, Ada language has many features that would simplify embedded software development.

Java is another language used for embedded systems programming. It primarily finds usage in high-end mobile phones as it offers portability across systems and is also useful for

browsing applications. Java programs require Java Virtual Machine (JVM), which consume lot of resources. Hence it is not used for smaller embedded devices. Dynamic C and B# are some proprietary languages which are also being used in embedded applications. Efficient embedded C programs must be kept small and efficient; they must be optimized for code speed and code size. Good understanding of processor architecture embedded C programming and debugging tools facilitate this.

Difference between C and embedded C:

Though C and embedded C appear different and are used in different contexts, they have more similarities than the differences. Most of the constructs are same; the difference lies in their applications.

C is used for desktop computers, while embedded C is for microcontroller based applications. Accordingly, C has the luxury to use resources of a desktop PC like memory, OS, etc. While programming on desktop systems, we need not bother about memory. However, embedded C has to use with the limited resources (RAM, ROM, I/Os) on an embedded processor. Thus, program code must fit into the available program memory. If code exceeds the limit, the system is likely to crash.

Compilers for C (ANSI C) typically generate OS dependant executables. Embedded C requires compilers to create files to be downloaded to the microcontrollers/microprocessors where it needs to run. Embedded compilers give access to all resources which is not provided in compilers for desktop computer applications. Embedded systems often have the real-time constraints, which is usually not there with desktop computer applications.

Embedded systems often do not have a console, which is available in case of desktop applications. So, what basically is different while programming with embedded C is the mindset; for embedded applications, we need to optimally use the resources, make the program code efficient, and satisfy real time constraints, if any. All this is done using the basic constructs, syntaxes, and function libraries of 'C'.

6.3.2 Keil C51 C Compilers

- Direct C51 to generate a listing file
- Define manifest constants on the command line
- Control the amount of information included in the object file
- Specify the level of optimization to use
- Specify the memory models

Specify the memory space for variables The Keil C51 C Compiler for the 8051 microcontroller is the most popular 8051 C compiler in the world. It provides more features than any other 8051 C compiler available today.

The C51 Compiler allows you to write 8051 microcontroller applications in C that, once compiled, have the efficiency and speed of assembly language. Language extensions in the C51 Compiler give you full access to all resources of the 8051.

The C51 Compiler translates C source files into reloadable object modules which contain full symbolic information for debugging with the μ Vision Debugger or an in-circuit emulator. In addition to the object file, the compiler generates a listing file which may optionally include symbol table and cross reference information.

6.4 SOFTWARE DESCRIPTION

6.4.1 PROTEUS

Proteus PCB design electronic circuits can computer-aided design and circuit boards are designed.

6.4.2 ISIS (Intelligent Schematic Input System)

The ISIS Intelligent Schematic Input System (Intelligent Switching input system), is the environment for the design and simulation of electronic circuits. The component library includes claims more than 10,000 circuit components with 6000 Prospice Simulations models. Own components can be created and added to the library.

ISIS includes a base VSM engine with support for the following functions:

- DC / AC voltmeter and ammeter, oscilloscopes, logic analyzers
- Analog signal generators, digital pattern generator
- Timer functions, protocol analyzers (including RS232, I2C, SPI)

6.4.3 VSM (Virtual System Modeling)

The VSM Virtual System Modeling provides a graphical SPICE circuit simulation and animation directly in the ISIS environment. The SPICE simulator is based on the Berkeley SPICE3F5 model.

It can microprocessor-based systems can be simulated. With the VSM engine can interact during the simulation directly with the circuit. Changes of buttons, switches or potentiometers are queried in real-time and LED indicators, LCD displays, "Hot / Cold" Wires displayed.

Proteus 7.0 is a Virtual System Modeling that combines circuit simulation, animated components and microprocessor models to co-simulate the complete microcontroller based designs. This is the perfect tool for engineers to test their microcontroller designs before constructing a physical prototype in real time. This program allows users to interact with the design using on-screen indicators and/or LED and LCD displays and, if attached to the PC, switches and buttons.

One of the main components of Proteus 7.0 is the Circuit Simulation -- a product that uses a SPICE3f5 analogue simulator kernel combined with an event-driven digital simulator that allow users to utilize any SPICE model by any manufacturer. Proteus VSM comes with extensive debugging features, including breakpoints, single stepping and variable display for a neat design prior to hardware prototyping.

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In summary, Proteus 7.0 is the program to use when you want to simulate the interaction between software running on a microcontroller and any analog or digital electronic device connected to it.

6.5 FEATURES OF PHP

PHP: Hypertext Preprocessor (the name is a recursive acronym) is a widely used, general-purpose scripting language that was originally designed for web development to produce dynamic web pages. For this purpose, PHP code is embedded into the HTML source document and interpreted by a web server with a PHP processor module, which generates the web page document. As a general-purpose programming language, PHP code is processed by an interpreter application in command-line mode performing desired operating system operations and producing program output on its standard output channel. It may also function as a graphical application. PHP is available as a processor for most modern web servers and as standalone interpreter on most operating systems and computing platforms.

PHP was originally created by Rasmus Lerdorf in 1995[1] and has been in continuous development ever since. The main implementation of PHP is now produced by The PHP Group and serves as the de facto standard for PHP as there is no formal specification. PHP is free software released under the PHP License, which is incompatible with the GNU General Public License (GPL) because restrictions exist regarding the use of the term PHP.

Hypertext refers to files linked together using hyperlinks, such as HTML (Hyper Text Markup Language) files. Preprocessing is executing instructions that modify the output. Below is a demonstration of the difference between HTML and PHP files.

Accessing an HTML Page

1. Your browser sends a request to that web page's server (computer) for the file (HTML or image) you wish to view.
2. The web server (computer) sends the file requested back to your computer.
3. Your browser displays the file appropriately.
4. If you request a PHP file (ends with ".php"), the server handles it differently.

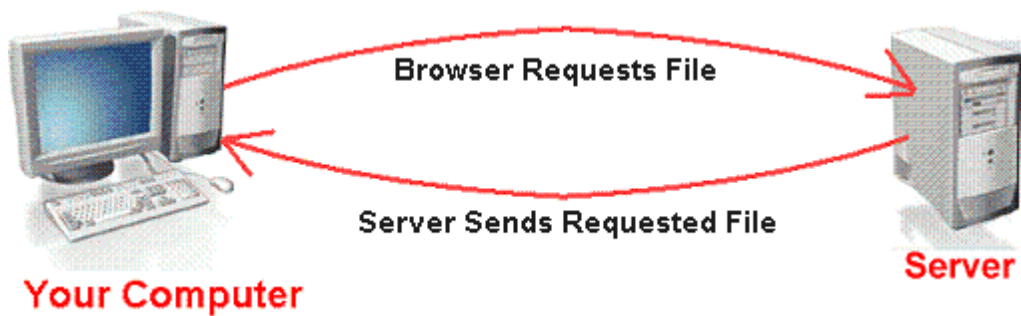


Fig 6.1 Accessing an HTML Pages

Accessing a PHP Page

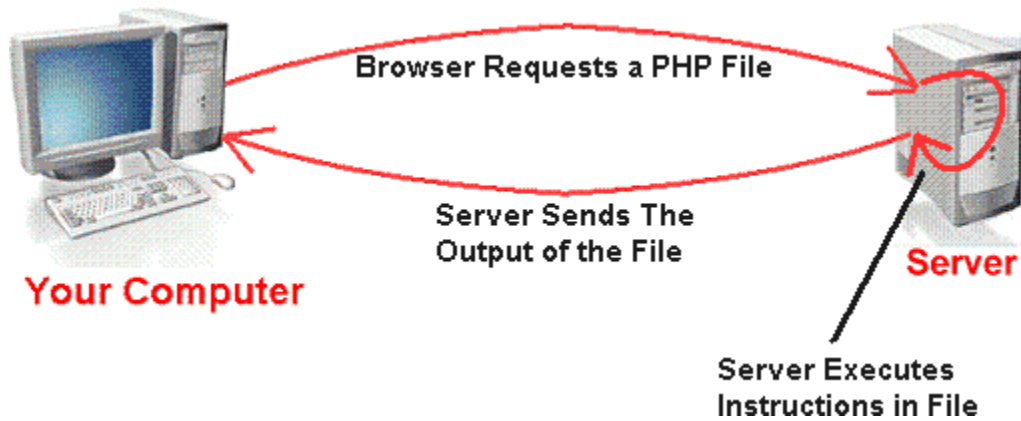


Fig 6.2 Accessing a PHP Pages

1. Your browser sends a request to that web page's server for the PHP file you wish to view.
2. The web server calls PHP to interpret and perform the operations called for in the PHP script.
3. The web server sends the output of the PHP program back to your computer.
4. Your browser displays the output appropriately.

PHP originally stood for personal home page. Its development began in 1994 when the Danish/Greenlandic programmer Rasmus Lerdorf initially created a set of Perl scripts he called 'Personal Home Page Tools' to maintain his personal homepage, including tasks such as displaying his résumé and recording how much traffic his page was receiving. He rewrote these scripts as C programming language Common Gateway Interface (CGI) binaries, extending them to add the ability to work with web forms and to communicate with databases and called this implementation 'Personal Home Page/Forms Interpreter' or PHP/FI. PHP/FI could be used to build simple, dynamic web applications. Lerdorf released PHP/FI as 'Personal Home Page Tools (PHP Tools) version 1.0' publicly on June 8, 1995, to accelerate bug location and improve the code. This release already had the basic functionality that PHP has today.

This included Perl-like variables, form handling, and the ability to embed HTML. The syntax was similar to Perl but was more limited and simpler, although less consistent. A development team began to form and, after months of work and beta testing, officially released PHP/FI 2 in November 1997. A new major version has been under development alongside PHP 5 for several years. This version was originally planned to be released as PHP 6 as a result of its significant changes, which included plans for full Unicode support. However, Unicode support took developers much longer to implement than originally thought, and the decision was made in March 2010[13] to move the project to a branch, with features still under development moved to trunk.

Changes in the new code include the removal of register globals, magic quotes, and safe mode. The reason for the removals was that register_globals had given way to security holes, and the use of magic quotes had an unpredictable nature, and was best avoided. Instead, to escape characters, magic quotes may be replaced with the addslashes() function, or more appropriately an escape mechanism specific to the database vendor itself like mysql_real_escape_string() for MySQL. Functions that will be removed in future versions and have been deprecated in PHP 5.3 will produce a warning if used.

PHP currently does not have native support for Unicode or multibyte strings; Unicode support is under development for a future version of PHP and will allow strings as well as class, method, and function names to contain non-ASCII characters.

PHP interpreters are available on both 32-bit and 64-bit operating systems, but on Microsoft Windows the only official distribution is a 32-bit implementation, requiring Windows 32-bit compatibility mode while using Internet Information Services (IIS) on a 64-bit Windows platform. As of PHP 5.3.0, experimental 64-bit versions are available for MS Windows.

Security

The PHP interpreter only executes PHP code within its delimiters. Anything outside its delimiters is not processed by PHP (although non-PHP text is still subject to control structures described within PHP code). The most common delimiters are `<?php` to open and `?>` to close PHP sections. `<script language="php">` and `</script>` delimiters are also available, as are the shortened forms `<?` or `<?=` (which is used to echo back a string or variable) and `?>` as well as ASP-style short forms `<%` or `<%=` and `%>`. While short delimiters are used, they make script files less portable as support for them can be disabled in the PHP configuration, and so they are discouraged. The purpose of all these delimiters is to separate PHP code from non-PHP code, including HTML.

The first form of delimiters, `<?php` and `?>`, in XHTML and other XML documents, creates correctly formed XML 'processing instructions'. This means that the resulting mixture of PHP code and other markup in the server-side file is itself well-formed XML.

Variables are prefixed with a dollar symbol and a type does not need to be specified in advance. Unlike function and class names, variable names are case sensitive. Both double-quoted (") and heredoc strings allow the ability to embed a variable's value into the string. PHP treats newlines as whitespace in the manner of a free-form language (except when inside string quotes), and statements are terminated by a semicolon. PHP has three types of comment syntax: `/* */` marks block and inline comments; `//` as well as `#` are used for one-line comments. The echo statement is one of several facilities PHP provides to output text (e.g. to a web browser).

In terms of keywords and language syntax, PHP is similar to most high level languages that follow the C style syntax. if conditions, for and while loops, and function returns are similar in syntax to languages such as C, C++, Java and Perl.

6.6 Benefit of PHP

Because the server does processing, the output of PHP files changes when its input changes. For example, most of the pages on the Horticulture site have only two (2) PHP commands:

1. Include the header file that defines the links on the left, the banner, and the quick links at the top.
2. Include the footer file that displays the mission statement and Horticulture contact information.

Because including the files is performed every time the PHP file is accessed, when the header/footer files change, the new content will be immediately updated. In other words, if you add a new link, every page that includes the header will immediately display the new link.

PHP has become the most popular Web programming language not only because it is free. PHP is a full-fledged programming language (unlike HTML for example, which is more of a presentation means) and many complex applications can be written in it. Another benefit of applications written in PHP is that they are fast and if written properly, they could be pretty secure. There are also tons of ready PHP scripts and functions, which you can customize to your liking and use in your PHP applications.

Web development is becoming a more prosperous industry lately. Since the Internet and the computer wave in general are becoming quite lucrative, web development is becoming a booming industry in which everyone wants to be a part. However, it is also a very competitive industry since there are many professionals that are quite adept at programming. Therefore,

learning proper PHP development strategies is beneficial. If you already have prior knowledge of computer language and coding, PHP should come as a second nature to you. The fact that it can be used in its most primal forms for basic programming as well as incredibly advanced programming only adds to its possible potential for programmers.

Before building a website you need to know which language you are going to use in a professional looking website. PHP is one of the best and easy to use programming languages as it can be run on any operating system. PHP is a free language so that is the huge advantage of this language. For handling database connections, formatting dates, editing strings, handling emails and all PHP can be very useful. It can be easily extended for some specific functions that you would like to add in your website. Reliability of this language is extraordinary as PHP already runs on millions of servers around the world, which means that it's powerful enough for even the most demanding situations. It provides web developers much more liberation in creating websites with some of the outstanding features and they can use regular elements frequently. PHP can be very much successful for creating Dynamic Websites. PHP programmers with the use of open source codes benefit from the flexibility of editing, modifying and updating the source code when there is a mandatory.

PHP is based on C++ programming language and the syntax used in PHP is fairly similar to C, C++. There is a huge community of developers who still believe that C/C++ is still the best programming language. For every website to get reasonable progress it can be used. Content Management Systems such as Joomla, WordPress etc. here PHP and MySQL are very helpful in successful CMS running. There are so many IT companies which provide best quality PHP web development work from India. The reason for outsourcing PHP development to India is that it is very cost effective with better quality. In the professional field of Web and software development services we have achieved great amount of victory with skilled and experienced PHP programmers.

PHP has been supported by almost every hosting company. PHP can be the best choice to run an application on Linux based hosting platforms. PHP is a server side scripting language originally designed to build dynamic websites. Modern web 2.0 applications are largely

characterized by mashups and desktop style user interfaces. php is an excellent choice for interacting with other websites and providing rich user experience. Simple php commands like curl or fopen allow you to grab data from other websites with relative ease. Php works well with javascript so you can provide your end users with modern, responsive interfaces that are way beyond the old static interfaces of days past.

6.7 BACKEND SOFTWARE

6.7.1 FEATURES OF MYSQL

6.7.1.1 MySQL Introduction

The MySQL® database has become the world's most popular open source database because of its consistent fast performance, high reliability and ease of use. It's used on every continent -- Yes, even Antarctica! -- by individual Web developers as well as many of the world's largest and fastest-growing organizations to save time and money powering their high-volume Web sites, business-critical systems and packaged software -- including industry leaders such as Yahoo!, Alcatel-Lucent, Google, Nokia, YouTube, and Zappos.com.

Not only is MySQL the world's most popular open source database, it's also become the database of choice for a new generation of applications built on the LAMP stack (Linux, Apache, MySQL, PHP / Perl / Python.) MySQL runs on more than 20 platforms including Linux, Windows, Mac OS, Solaris, HP-UX, IBM AIX, giving you the kind of flexibility that puts you in control. Whether you're new to database technology or an experienced developer or DBA, MySQL offers a comprehensive range of certified software, support, training and consulting to make you successful. MySQL can be built and installed manually from source code, but this can be tedious so it is more commonly installed from a binary package unless special customizations are required. On most Linux distributions the package management system can download and install MySQL with minimal effort, though further configuration is often required to adjust security and optimization settings.

Though MySQL began as a low-end alternative to more powerful proprietary databases, it has gradually evolved to support higher-scale needs as well. It is still most commonly used in small to medium scale single-server deployments, either as a component in a LAMP based web application or as a standalone database server. Much of MySQL's appeal originates in its relative simplicity and ease of use, which is enabled by an ecosystem of open source tools such as phpMyAdmin. In the medium range, MySQL can be scaled by deploying it on more powerful hardware, such as a multi-processor server with gigabytes of memory.

There are however limits to how far performance can scale on a single server, so on larger scales, multi-server MySQL deployments are required to provide improved performance and reliability. A typical high-end configuration can include a powerful master database which handles data write operations and is replicated to multiple slaves that handle all read operations.[18] The master server synchronizes continually with its slaves so in the event of failure a slave can be promoted to become the new master, minimizing downtime. Further improvements in performance can be achieved by caching the results from database queries in memory using memcached, or breaking down a database into smaller chunks called shards which can be spread across a number of distributed server clusters.

CODING :

```
#include <Servo.h>

// ----- PINS -----
#define IR_SENSOR A3
#define BUZZER 4
#define METAL_PIN A2
#define MOISTURE_PIN A0
#define RELAY_PIN 7

Servo wasteServo; // sorting servo
Servo lidServo;   // NEW: open/close lid servo

// ----- VARIABLES -----
int moistureValue;
int metalDetected;
int currentAngle = 0;

// ----- LID ANGLES -----
#define LID_OPEN 180
#define LID_CLOSE 0

// ----- SERVO SMOOTH FUNCTION -----
void moveServoSmooth(int toAngle) {

    if (currentAngle < toAngle) {
        for (int i = currentAngle; i <= toAngle; i++) {
            wasteServo.write(i);
            delay(5);
        }
    } else {
        for (int i = currentAngle; i >= toAngle; i--) {
            wasteServo.write(i);
            delay(5);
        }
    }

    currentAngle = toAngle;
}
```

```

// ----- LID CONTROL -----
void openLid() {
  for (int i = LID_CLOSE; i <= LID_OPEN; i++) {
    lidServo.write(i);
    delay(5);
  }
}

void closeLid() {
  for (int i = LID_OPEN; i >= LID_CLOSE; i--) {
    lidServo.write(i);
    delay(5);
  }
}

void setup() {
  Serial.begin(9600);

  pinMode(IR_SENSOR, INPUT);
  pinMode(BUZZER, OUTPUT);
  pinMode(METAL_PIN, INPUT);
  pinMode(RELAY_PIN, OUTPUT);

  wasteServo.attach(10);
  lidServo.attach(9); // NEW SERVO PIN

  wasteServo.write(0);
  lidServo.write(LID_CLOSE);

  currentAngle = 0;

  digitalWrite(BUZZER, LOW);
  digitalWrite(RELAY_PIN, LOW);

  Serial.println("SMART WASTE SYSTEM READY");
}

```

```

void loop() {

  int objectDetected = digitalRead(IR_SENSOR);

  // ----- WAIT FOR OBJECT -----
  if (objectDetected == HIGH) {
    digitalWrite(BUZZER, LOW);
    delay(200);
    return;
  }

  // ----- OBJECT DETECTED -----
  Serial.println("OBJECT DETECTED");

  digitalWrite(BUZZER, HIGH);
  delay(300);
  digitalWrite(BUZZER, LOW);

  delay(300);

  // ----- READ SENSORS -----
  moistureValue = analogRead(MOISTURE_PIN);
  metalDetected = digitalRead(METAL_PIN);

  Serial.print("Moisture: ");
  Serial.print(moistureValue);
  Serial.print(" | Metal: ");
  Serial.println(metalDetected);

  // ----- FOOD WASTE -----
  if (moistureValue < 800) {
    Serial.println("FOOD WASTE DETECTED");

    moveServoSmooth(180);
    delay(1500);
    openLid();    // OPEN DROP
    delay(1500);  // wait for waste to fall
    closeLid();   // CLOSE
    delay(1500);
  }
}

```

```

    // ----- RELAY MOTOR -----
    digitalWrite(RELAY_PIN, HIGH);
    delay(3500);
    digitalWrite(RELAY_PIN, LOW);

    delay(1000);
}

// ----- METAL WASTE -----
else if (metalDetected == LOW) {
    Serial.println("METAL WASTE DETECTED");

    moveServoSmooth(90);
    delay(1500);
    openLid();
    delay(1500);
    closeLid();
    delay(1500);
    // ----- RELAY MOTOR -----
    digitalWrite(RELAY_PIN, HIGH);
    delay(3500);
    digitalWrite(RELAY_PIN, LOW);

    delay(1000);
}

// ----- PLASTIC WASTE -----
else {
    Serial.println("PLASTIC WASTE DETECTED");

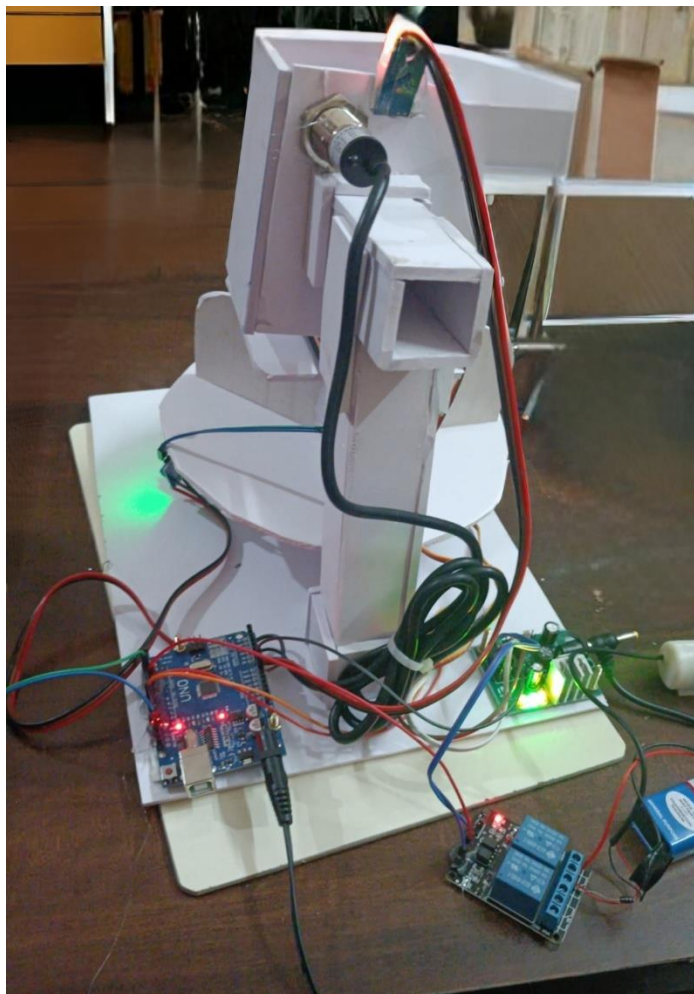
    moveServoSmooth(0);
    delay(1500);
    openLid();
    delay(1500);
    closeLid();
    delay(1500);

    // ----- RELAY MOTOR -----
    digitalWrite(RELAY_PIN, HIGH);
    delay(3500);
    digitalWrite(RELAY_PIN, LOW);

    delay(1000);
}
}

```

SCREEN SHOT



CONCLUSION

The Smart Waste Management System using IoT and Artificial Intelligence provides an effective and modern solution to the challenges faced in traditional waste management practices. By enabling proper segregation of waste into biodegradable and non-biodegradable categories at the source level, the system significantly improves recycling efficiency and reduces environmental pollution. The integration of smart sensors, microcontrollers, and IoT communication modules ensures real-time monitoring of waste bins, preventing overflow and maintaining cleanliness in residential and public areas. The automated segregation process minimizes human effort and reduces health risks associated with manual waste handling. Additionally, the use of a mobile or web application enhances user interaction and allows authorities to manage waste collection efficiently. Furthermore, data analytics and predictive techniques help optimize waste collection schedules and routes, leading to reduced operational costs and improved resource utilization. The implementation of this system demonstrates the potential of combining IoT and AI technologies to create sustainable and scalable solutions for smart cities. In conclusion, the proposed system not only addresses the existing inefficiencies in waste management but also promotes environmental sustainability, public hygiene, and technological advancement. It serves as a reliable step toward building cleaner, greener, and smarter communities.

REFERENCES

- [1]. Shyam, G.K., Manvi, S.S., Bharti, P.: Smart waste management using Internet-of-Things (IoT). In: 2017 2nd International Conference on Computing and Communications Technologies (ICCCT), Chennai, pp. 199–203 (2017)
- [2]. Bharadwaj, A.S., Rego, R., Chowdhury, A.: IoT based solid waste management system: a conceptual approach with an architectural solution as a smart city application. In: 2016 IEEE Annual India Conference (INDICON), Bangalore, pp. 1–6 (2016)
- [3]. Wijaya, A.S., Zainuddin, Z., Niswar, M.: Design a smart waste bin for smart waste management. In: 2017 5th International Conference on Instrumentation, Control, and Automation (ICA), Yogyakarta, pp. 62–66 (2017)
- [4]. Cloud-based Smart Waste Management for Smart Cities Mohammad Aazam, Marc StHillarie, Chung-Horng Lung, Ioannis Lambadaris 2016 IEEE 21st International Workshop on Computer Aided Modeling and Design of Communication Links and Networks (CAMAD)
- [5]. Automation of smart waste management using iot to support “swachh bharat abhiyan” – a practical approach 1 Bharadwaj B, 2 M Kumudha, 3 Gowri Chandra N, 4 Chaithra G, 2017 2nd International conference on computing and communications technology
- [6]. Smart Waste Management System using IoT Prof. S.A. Mahajan, Akshay Kokane, Apoorva Shewale, Mrunaya Shinde , Shivani Ingale, [Vol4, Issue-4, Apr- 2017] ISSN: 2349-6495(P) | 2456- 1908(O)
- [7]. Smart Waste Management using Internet-ofThings (IoT) Gopal Kirshna Shyam¹, Sunilkumar S. Manvi², Priyanka Bharti³ 2017 2nd International conference on computing and communications technologies (ICCCT)

- [8]. IoT based solid waste management system for smart city Krishna Nirde 1, Prashant S. Mulay 2, Uttam M.Chaskar 3 International Conference on Intelligent Computing and Control Systems ICICCS 2017
- [9]. Smart Garbage Monitoring and Clearance System using Internet of Things S. Vinoth Kumar¹, T. Senthil Kumaran², A. Krishna Kumar³ and Mahantesh Mathapati⁴ 2017 IEEE International Conference on Smart Technologies and Management 2-4
- [10]. Efficient IOT Based Smart Bin for Clean Environment Murugaanandam. S, Ganapathy. V and Balaji. R 2018 International Conference on Communication and signal processing (ICCSP)
- [11]. Smart Waste Management using Internet-ofThings (IoT) Chitluri Sai Srikanth, R. Anitha 2019
- [12]. IoT-based Intelligent Waste Bin Moath Awarded , Anees Bashir, Tarig Faisal, Irfan Ahmad, and Muhammad Khalil Shahid, 2019 advances in Science and engineering technology international conferences (ASET)
- [13]. Smart waste management system 1 Shyamala S.C, 2Kunjan Sindhe, 3Vishwanth Muddy, 4Chitra C N Assistant Professors, International Journal of Scientific Development and Research (IJSDR)
- [14]. KristýnaRybová, Jan Slavík, “Smart cities and ageing population– Implications for waste management in the Czech Republic “ -IEEE 2016.

PATENT

Mr.J. SAMPATHKUMAR, Dr.M.Suganthi, Mr.D.Balaji, "3-D based brain tumor detection using (DL) Deep learning and image processing Approach", 2021

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